

Distinct zeros of fixed Dedekind zeta functions

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Abstract

Let $K/\mathbb{Q}$ be a fixed number field of degree n , let M be its normal closure, and put $G = \text{Gal}(M/\mathbb{Q})$, $H = \text{Gal}(M/K)$, and $r = |H \backslash G/H|$. Writing $N_K(T)$ for the number of nontrivial zeros ρ of ζ_K with $0 < \text{Im } \rho \leq T$, counted with multiplicity, and $D_K(T)$ for the number of distinct complex zero locations, we prove unconditionally that

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq \frac{2n+1-C_{n,r}}{n(n+1)}, \quad C_{n,r} = \frac{r}{2n} + \sqrt{\frac{r}{2}} \cot\left(\frac{1}{n}\sqrt{\frac{r}{2}}\right).$$

The resulting degree-only bounds in degrees two, three, and four are 44.49%, 30.57%, and 23.34%. Our proof combines an integer-threshold extension of Lamzouri's Hilbert-space certificate with a strict-support pair formula for the actual complex zeros of ζ_K ; its arithmetic coefficient is the permutation rank r . A projection-and-packing argument gives a further explicit positive improvement when $n \geq 2$. No GRH or global Artin-holomorphy hypothesis is used; the field is fixed throughout.

Keywords. Dedekind zeta function; distinct zeros; pair correlation; multiplicity; Hilbert-space method; Chebotarev theorem.

1 Introduction

The factorization of a Dedekind zeta function makes multiplicity a substantive issue. For example, if $K/\mathbb{Q}$ is quadratic, then $\zeta_K(s) = \zeta(s)L(s, \chi_K)$, and a common zero of the two factors is a multiple zero of the product. Thus information about simple zeros of individual factors does not by itself give simple zeros of ζ_K . We ask instead for an unconditional lower bound for the number of distinct zero locations of the product.

Throughout, $K/\mathbb{Q}$ is fixed and $T \rightarrow \infty$. Let $\mathcal{Z}_K(T)$ be the set of distinct nontrivial zeros $\rho = \beta + i\gamma$ of ζ_K with $0 < \beta < 1$ and $0 < \gamma \leq T$, and let $m_\rho = \text{ord}_\rho \zeta_K$. We write

$$N_K(T) = \sum_{\rho \in \mathcal{Z}_K(T)} m_\rho, \quad D_K(T) = |\mathcal{Z}_K(T)|.$$

Thus $D_K(T)$ counts distinct complex locations, not merely ordinates, and $N_K(T)$ counts zeros with analytic multiplicity. Real zeros are excluded from both counts.

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1.1 Main theorem

Let $n = [K : \mathbb{Q}]$, let M be the normal closure of $K/\mathbb{Q}$, and set

$$G = \text{Gal}(M/\mathbb{Q}), \quad H = \text{Gal}(M/K), \quad r = |H \backslash G/H|.$$

The integer r is the permutation rank of the transitive action of G on the embeddings of K into M : equivalently, it is the number of H -orbits on G/H . Hence $1 \leq r \leq n$, with $r = n$ exactly when H is normal (equivalently, $K/\mathbb{Q}$ is Galois), and $r = 2$ exactly when G acts doubly transitively on G/H . Put

$$C_{n,r} = \frac{r}{2n} + \sqrt{\frac{r}{2}} \cot\left(\frac{1}{n}\sqrt{\frac{r}{2}}\right).$$

Theorem 1. *For every fixed number field $K/\mathbb{Q}$, without GRH or the Artin holomorphy conjecture,*

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq b_{n,r} := \frac{2n+1-C_{n,r}}{n(n+1)}. \quad (1)$$

In particular,

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq \frac{1}{n} - \frac{r}{3n^2(n+1)} \geq \frac{3n+2}{3n(n+1)}. \quad (2)$$

The bound in (1) is strictly stronger than the first bound in (2). If $n \geq 2$, Theorem 11 gives the further explicit improvement $b_{n,r} + \epsilon_{n,r}$ with $\epsilon_{n,r} > 0$.

For fixed n , the constant $C_{n,r}$ is strictly increasing in r . Consequently $b_{n,r} \geq b_{n,n}$, with strict inequality when $r < n$. The following specializations therefore hold for every field of the indicated degree.

Table 1: Degree-only specializations of (1); the geometric increment is not included.

Degree	Distinct fraction	Percentage
2	0.44491871304792468...	44.4918713...%
3	0.30571236760745205...	30.5712367...%
4	0.23340361496350142...	23.3403614...%

For a non-Galois cubic field the permutation rank is $r = 2$, and (1) gives $b_{3,2} = 0.3148841358\dots$. A doubly transitive quartic field also has $r = 2$, giving $b_{4,2} = 0.2416841317\dots$. When $K = \mathbb{Q}$, the theorem recovers Lamzouri's distinct-zero constant [11, Theorem 1.1].

1.2 Method and context

The proof combines three ingredients. First, Lamzouri's Hilbert-space construction [11], applied to $z_\rho = i(\rho - 1/2) \log T/(2\pi)$, gives for every integer $k \geq 1$

$$D \geq \frac{(2k+1)N - E}{k(k+1)}.$$

The Dedekind energy is of size $nN_K(T)$, so the relevant threshold is $k = n$. Second, an unconditional pair formula for the actual complex zeros has main term

$$N_K(T) \left(nf(0) + \frac{r}{n} \int_{-1}^1 |u| f(u) du \right).$$

The coefficient r follows from Chebotarev and $|G|^{-1} \sum_{g \in G} P(g)^2 = r$ for the permutation character P . Third, weight removal reduces the problem to minimizing

$$J_{n,r}(p) = n \int_I p^2 + \frac{r}{n} \iint_{I^2} |x - y| p(x) p(y) dx dy, \quad \int_I p = 1,$$

whose cosine minimizer has value $C_{n,r}$. A projection-and-packing argument yields the strict improvement in Theorem 11.

The complex-zero framework follows [4, 5, 6], and the variational constant is related to generalized Montgomery–Taylor theory [12, 8]. Earlier Dedekind pair-correlation estimates in [2, 9, 15] assume GRH. Sections 2–4 prove Theorem 1, Section 5 proves the refinement, and Section 6 gives a brief comparison.

2 The integer-threshold Hilbert certificate

Let $\eta \in L^2(\mathbb{R})$ be real, even, and compactly supported, with $\int_{\mathbb{R}} \eta(u)^2 du = 1$. For compactly supported $f \in L^1(\mathbb{R})$, use the entire Fourier transform

$$\widehat{f}(z) = \int_{\mathbb{R}} f(u) e^{-2\pi i z u} du \quad (z \in \mathbb{C}), \quad K_{\eta}(z) = \widehat{\eta^2}(z).$$

Compact support makes this extension entire. In particular, K_{η} is even, is real on $\mathbb{R}$, and satisfies $K_{\eta}(0) = 1$. Let Z be a finite nonempty multiset with support $\Omega \subset \mathbb{C}$ and positive integer multiplicities m_z . Assume that Z is invariant under complex conjugation: $\bar{z} \in \Omega$ and $m_{\bar{z}} = m_z$ for every $z \in \Omega$. Define

$$N = \sum_{z \in \Omega} m_z, \quad D = |\Omega|, \quad N_C = \sum_{z \in \Omega \setminus \mathbb{R}} m_z,$$

and

$$E = \sum_{z, w \in \Omega} m_z m_w K_{\eta}(z - w)^2.$$

This is an ordered double sum with multiplicities. The square is an ordinary square, not an absolute square; individual summands need not be real. The proof below shows that the total E is real and positive. No assertion about zeta functions or an asymptotic limit is needed in this finite lemma.

Lemma 2. *For every integer $k \geq 1$,*

$$D \geq \frac{(2k+1)N - E}{k(k+1)}. \quad (3)$$

Moreover, the threshold defect satisfies

$$\Delta_k := E - (2k+1)N + k(k+1)D \geq N_C. \quad (4)$$

Proof. *The real tensor and its energy.* Use Lamzouri’s functions [11, Proposition 2.1]

$$f_z(u) = \eta(u) e^{-2\pi i z u}, \quad g_z = (f_z + f_{\bar{z}})/2, \quad h_z = (f_z - f_{\bar{z}})/(2i).$$

Compact support ensures that these functions belong to $L^2(\mathbb{R}; \mathbb{C})$ for every fixed $z \in \mathbb{C}$. On this space we use the inner product, linear in the first variable,

$$\langle \Phi, \Psi \rangle = \int_{\mathbb{R}} \Phi(u) \overline{\Psi(u)} du, \quad \langle f_z, f_w \rangle = K_{\eta}(z - \bar{w}).$$

The conjugate-linear isometric involution $J\Phi(u) = \overline{\Phi(-u)}$ has the closed real fixed space $\mathcal{H}_0 = \{\Phi : J\Phi = \Phi\}$, with real scalars. Changing u to $-u$ shows that the inner product of two elements of $\mathcal{H}_0$ is real, so this is a real Hilbert space. Since $Jf_z = f_{\bar{z}}$, the vectors f_x for real x , and all g_z, h_z , belong to $\mathcal{H}_0$. Furthermore,

$$\|f_x\|^2 = 1, \quad \|g_z\|^2 - \|h_z\|^2 = \operatorname{Re}\langle f_z, f_{\bar{z}} \rangle = 1.$$

For $a, b \in \mathcal{H}_0$, identify the real tensor $a \otimes b$ with the operator $v \mapsto \langle v, b \rangle a$. Let $\mathcal{P}$ contain one representative from each nonreal conjugate pair. Define the real symmetric finite-rank operator

$$F = \sum_{x \in \Omega \cap \mathbb{R}} m_x f_x \otimes f_x + 2 \sum_{z \in \mathcal{P}} m_z (g_z \otimes g_z - h_z \otimes h_z).$$

In the complexified tensor product, $f_z = g_z + ih_z$ gives $F = \sum_{z \in \Omega} m_z f_z \otimes f_z$. Here tensor products are complex bilinear; this is *not* a sum of positive Hermitian rank-one operators. The tensor inner product satisfies $\langle a \otimes b, c \otimes d \rangle = \langle a, c \rangle \langle b, d \rangle$. Consequently its norm, which agrees with the Hilbert–Schmidt norm of the real operator, satisfies

$$\begin{aligned} \|F\| * \operatorname{HS}^2 &= \sum_{z, w \in \Omega} m_z m_w K_\eta(z - \bar{w})^2 \\ &= \sum_{z, w \in \Omega} m_z m_w K_\eta(z - w)^2 = E. \end{aligned}$$

The second equality reindexes $w \mapsto \bar{w}$ and uses equality of the two multiplicities. The real decomposition also gives

$$\operatorname{tr} F = \sum_{x \in \Omega \cap \mathbb{R}} m_x + 2 \sum_{z \in \mathcal{P}} m_z (\|g_z\|^2 - \|h_z\|^2) = N.$$

In particular $F \neq 0$ and $E > 0$, because $N > 0$.

The three-block bound and the nonreal defect. Partition real locations into $m_x \leq k$ and $m_x \geq k + 1$; denote their counts by D_L, D_H and masses by M_L, M_H . Let $c_1 = |\mathcal{P}|$ count nonreal conjugate pairs and put $M_C = \sum_{z \in \mathcal{P}} m_z$. Thus $N = M_L + M_H + 2M_C$, $D = D_L + D_H + 2c_1$, and $N_C = 2M_C$. Take

$$\begin{aligned} U &= \operatorname{span}_{\mathbb{R}}(f_x : x \in \Omega \cap \mathbb{R}, m_x \geq k + 1; g_z : z \in \mathcal{P}), \\ V &= U + \operatorname{span}_{\mathbb{R}}(f_x : x \in \Omega \cap \mathbb{R}, m_x \leq k), \quad W = V + \operatorname{span}_{\mathbb{R}}(h_z : z \in \mathcal{P}). \end{aligned}$$

The range of F is contained in W , and F vanishes on $W^\perp$. Choose an orthonormal basis $(\psi_j)_{j=1}^{\dim W}$ adapted to $U \subseteq V \subseteq W$, omitting dependent spanning vectors during real Gram–Schmidt. Put $\alpha_j = \langle F\psi_j, \psi_j \rangle = \langle F, \psi_j \otimes \psi_j \rangle \in \mathbb{R}$. The tensors $\psi_j \otimes \psi_j$ are orthonormal. Bessel’s inequality and the trace identity give

$$E = \|F\| * \operatorname{HS}^2 \geq \sum_j \alpha_j^2, \quad \sum_j \alpha_j = N. \quad (5)$$

Let τ_U, τ_V, τ_W be the sums of the α_j over the basis vectors lying in $U, V \ominus U, W \ominus V$, respectively; these are the traces of F compressed to the three blocks, and $\tau_U + \tau_V + \tau_W = N$. If Q_V projects orthogonally onto the middle block $V \ominus U$, then

$$\tau_V = \sum_{x \in \Omega \cap \mathbb{R}}^* m_x \|Q_V f_x\|^2 - 2 \sum_{z \in \mathcal{P}} m_z \|Q_V h_z\|^2 \leq M_L.$$

Indeed, Q_V annihilates every high-multiplicity real vector and every g_z , while $\|Q_V f_x\| \leq \|f_x\| = 1$. For a basis vector in the last block, $\alpha_j = -2 \sum_{z \in \mathcal{P}} m_z \langle h_z, \psi_j \rangle^2 \leq 0$. Also $d_U := \dim U \leq D_H + c_1$ and $d_V := \dim(V \ominus U) \leq D_L$. Apply $a^2 \geq 2(k+1)a - (k+1)^2$ on the first block, $a^2 \geq 2ka - k^2$ on

the second, and $a^2 \geq 2(k+1)a$ on the last, where the latter uses $a \leq 0$. Summing, and then using $\tau_U + \tau_V + \tau_W = N$, gives

$$\begin{aligned} E &\geq 2(k+1)\tau_U - (k+1)^2 d_U + 2k\tau_V - k^2 d_V + 2(k+1)\tau_W \\ &= 2(k+1)N - (k+1)^2 d_U - k^2 d_V - 2\tau_V \\ &\geq 2(k+1)N - (k+1)^2 (D_H + c_1) - k^2 D_L - 2M_L. \end{aligned}$$

Subtracting $(2k+1)N - k(k+1)D$ therefore gives

$$\Delta_k \geq [M_H - (k+1)D_H] + [kD_L - M_L] + 2M_C + (k^2 - 1)c_1.$$

Both brackets are nonnegative by the multiplicity ranges, and $k^2 - 1 \geq 0$. The right-hand side is therefore at least N_C , proving both (3) and (4).

All arguments remain valid when blocks are empty or spanning vectors are dependent: dimensions can only decrease, and empty traces and sums are zero. $\square$

Choice of threshold and comparison with rank–trace. The case $k = 1$ in (3) recovers Lamzouri’s distinct-element inequality. Since $\dim W \leq D \leq N$, (5) and Cauchy–Schwarz give

$$N^2 \leq (\dim W) \sum_j \alpha_j^2 \leq DE.$$

Thus $D \geq N^2/E$ and $E/N \geq N/D \geq 1$. If $E/N \leq C$ for a real constant C , necessarily $C \geq 1$. The basic bound (3), with E replaced by CN , becomes $D/N \geq B_k(C)$, where

$$B_k(C) = \frac{2k+1-C}{k(k+1)}, \quad B_{k+1}(C) - B_k(C) = \frac{2(C-k-1)}{k(k+1)(k+2)}.$$

The sign of this difference shows that $k = \lfloor C \rfloor$ is a maximizer over integers $k \geq 1$. It is unique for nonintegral C ; at an integer $C = m \geq 2$ the two maximizers are $m-1$ and m . For $C = 1$ the unique admissible maximizer is $k = 1$. On $k \leq C \leq k+1$, comparison with the rank–trace consequence $D/N \geq 1/C$ gives

$$B_k(C) - \frac{1}{C} = \frac{(C-k)(k+1-C)}{Ck(k+1)} \geq 0.$$

The improvement is strict for nonintegral C and vanishes at integers when an optimal threshold is used.

3 An unconditional pair formula for every fixed field

Throughout this section the number field K is fixed, $n = [K : \mathbb{Q}]$, $d_K = |\text{disc } K|$, and (r_1, r_2) is its signature, so $n = r_1 + 2r_2$. Constants denoted by O_K are independent of all the displayed real variables in their stated ranges. They need not be uniform in the field. Unless a height restriction is displayed, a sum over $\rho = \beta + i\gamma$ runs over all nontrivial zeros of ζ_K , including real zeros. Every zero sum counts analytic multiplicity, and every pair sum is over ordered pairs, including the diagonal. Thus $N_K(T)$ counts $0 < \gamma \leq T$, with multiplicity, not both signs of the ordinate. For compactly supported integrable f , our Fourier convention and its entire extension are

$$\hat{f}(z) = \int_{\mathbb{R}} f(u) e^{-2\pi i u z} du, \quad z \in \mathbb{C}, \quad W(z) = \frac{4}{4 - z^2}.$$

Compact support justifies differentiation under the Fourier integral at every complex argument. The function W is meromorphic, but its poles at ± 2 are never reached by a zero difference, since $|\text{Re}(\rho - \rho')| < 1$.

Lemma 3 (Strict-support pair formula). *Let $0 < b < 1$ and let $f \in C_c^2(\mathbb{R})$ be real and even, with $\text{supp } f \subset [-b, b]$. Put $L = \log T$. As $T \rightarrow \infty$, without GRH or Artin holomorphy,*

$$\begin{aligned} & \sum_{0 < \gamma, \gamma' \leq T} \widehat{f}\left(i(\rho - \rho') \frac{L}{2\pi}\right) W(\rho - \rho') \\ &= N_K(T) \left(n f(0) + \frac{r}{n} \int_{-1}^1 |u| f(u) du + O_{K,b,f}(1/\log T) \right), \end{aligned} \quad (6)$$

where $r = |H \backslash G/H|$ as in Section 1.1. The zeros in the sum are their actual complex locations. The formula holds for arbitrary real T sufficiently large, including heights that are zero ordinates.

The proof of this lemma is completed in Proposition 10. The restriction $b < 1$ is used explicitly when the sharp-window error is integrated. The proof of the main theorem uses fixed smooth interior cutoffs and therefore requires no endpoint-support pair formula.

3.1 Completed zeta function and an unconditional local zero count

Write

$$Z_K(s) = \frac{\zeta'_K(s)}{\zeta_K(s)}, \quad -Z_K(s) = \sum_{m \geq 2} A_K(m) m^{-s} \quad (\text{Re } s > 1).$$

The Euler product gives $0 \leq A_K(m) \leq n\Lambda(m)$; its full local calculation is given in Proposition 5 below. Set

$$\begin{aligned} \Gamma_{\mathbb{R}}(s) &= \pi^{-s/2} \Gamma(s/2), & \Gamma_{\mathbb{C}}(s) &= 2(2\pi)^{-s} \Gamma(s), \\ \xi_K(s) &= s(s-1) d_K^{s/2} \Gamma_{\mathbb{R}}(s)^{r_1} \Gamma_{\mathbb{C}}(s)^{r_2} \zeta_K(s). \end{aligned}$$

The classical unconditional facts used here are that ξ_K is entire of order one, is nonzero at 0 and 1, has precisely the nontrivial zeros of ζ_K , all in $0 < \text{Re } s < 1$, and satisfies

$$\xi_K(s) = \xi_K(1-s), \quad \xi_K(\bar{s}) = \overline{\xi_K(s)}.$$

See [10, Section 2.1, equations (2.1)–(2.2)] for the completed function, its order, and its functional equation, and [9, Section 2.1] for the trivial-zero multiplicities recorded in Proposition 6 below. The gamma duplication formula identifies the gamma factors in [10] with the ones above; a constant normalization of ξ_K , such as the factor $1/2$ in [9], does not affect its logarithmic derivative. Only these unconditional completed-function facts are imported: in particular, we do not use the GRH-dependent pair formula in [9]. The local zero estimate below is proved before the global normalization, so the two estimates do not depend circularly on one another.

Lemma 4 (Local zero count). *For every real t ,*

$$U_K(t) := \sum_{\rho} \frac{2 - \beta}{(2 - \beta)^2 + (t - \gamma)^2} = \frac{n}{2} \log(|t| + 2) + O_K(1), \quad (7)$$

$$\#\{\rho : |\gamma - t| \leq 1\} \ll_K \log(|t| + 2), \quad (8)$$

$$\sum_{\rho} \frac{1}{1 + (t - \gamma)^2} \ll_K \log(|t| + 2). \quad (9)$$

The series in these statements converge absolutely.

Proof. Order one and Jensen's formula give $\#\{\rho : |\rho| \leq R\} = O_K(R^{5/4})$, and consequently $\sum_\rho |\rho|^{-2} < \infty$. Hadamard factorization therefore gives

$$\frac{\xi'_K}{\xi_K}(s) = b_K + \sum_\rho \left(\frac{1}{s - \rho} + \frac{1}{\rho} \right),$$

with normally convergent combined terms off the zeros. Since $0 < \beta < 1$, the real parts of the two sums separately converge absolutely at any fixed s off the zeros. Hence

$$\operatorname{Re} \frac{\xi'_K}{\xi_K}(\sigma + it) = c_K + \sum_\rho \frac{\sigma - \beta}{(\sigma - \beta)^2 + (t - \gamma)^2}, \quad c_K = \operatorname{Re} b_K + \sum_\rho \operatorname{Re}(1/\rho).$$

In fact $c_K = 0$. The functional equation makes the logarithmic derivatives at 2 and -1 negatives. Reindexing the sum at -1 by $\rho \mapsto 1 - \bar{\rho}$ makes it the negative of the sum at 2. Adding the two displayed identities gives $2c_K = 0$.

At $s = 2 + it$, logarithmic differentiation of the completion now gives

$$U_K(t) = \operatorname{Re} \left(\frac{1}{s} + \frac{1}{s-1} + \frac{1}{2} \log d_K - \frac{r_1}{2} \log \pi - r_2 \log(2\pi) \right. \\ \left. + \frac{r_1}{2} \psi(s/2) + r_2 \psi(s) + Z_K(s) \right), \quad \psi = \Gamma'/\Gamma.$$

Here $|Z_K(2 + it)| \leq n \sum_m \Lambda(m) m^{-2}$ uniformly in t . Stirling's estimate for ψ on these fixed vertical lines, together with continuity for bounded t , proves (7). All summands are positive. If $|\gamma - t| \leq 1$, each is at least $1/5$, because $1 < 2 - \beta < 2$. This proves (8). Finally, for $a \in (1, 2)$,

$$\frac{a}{a^2 + y^2} \geq \frac{1}{4 + y^2} \geq \frac{1}{4(1 + y^2)}.$$

Summation and (7) prove (9). □

3.2 The arithmetic second moment

Proposition 5 (Permutation-rank coefficient). *Uniformly for $y \geq 2$,*

$$B_K(y) := \sum_{m \leq y} A_K(m)^2 = ry \log y + O_K(y). \quad (10)$$

For every rational prime p , including ramified primes, and every $j \geq 1$,

$$A_K(p^j) = \log p \sum_{\substack{\mathfrak{p}|p \\ f_{\mathfrak{p}}|j}} f_{\mathfrak{p}}, \quad 0 \leq A_K(m) \leq n\Lambda(m). \quad (11)$$

Here $f_{\mathfrak{p}}$ is the residue degree. All coefficients outside prime powers vanish.

Proof. The local Euler factor at p is $\prod_{\mathfrak{p}|p} (1 - p^{-f_{\mathfrak{p}}s})^{-1}$. Its negative logarithmic derivative is

$$\sum_{\mathfrak{p}|p} f_{\mathfrak{p}} \log p \sum_{k \geq 1} p^{-kf_{\mathfrak{p}}s}.$$

This proves the exact coefficient formula. Since $\sum_{\mathfrak{p}|p} e_{\mathfrak{p}} f_{\mathfrak{p}} = n$ and $e_{\mathfrak{p}} \geq 1$, it also proves the bound in (11) without a ramification restriction.

Let M be the normal closure, $G = \text{Gal}(M/\mathbb{Q})$, $H = \text{Gal}(M/K)$, and $\Omega = G/H$. Write $P(g) = \#\text{Fix}_\Omega(g)$. At a prime unramified in M , the cycle lengths of Frobenius on Ω are the residue degrees of the primes of K above p . Consequently

$$A_K(p) = P(\text{Frob} * p) \log p.$$

Burnside's lemma, applied to the diagonal action on Ω^2 , gives

$$\frac{1}{|G|} \sum *g \in GP(g)^2 = \#(G \backslash \Omega^2) = \#(H \backslash G/H) = r. \quad (12)$$

The second equality follows by moving the first coordinate to H ; its stabilizer is H , acting on the second coordinate by left multiplication.

For a conjugacy class $C \subset G$, let $\pi_C(y)$ count the rational primes at most y , unramified in M , with Frobenius in C . The fixed-extension consequence of effective Chebotarev [16, Theorem 1.1] (or [14, Theorem 1.1]) is

$$\pi_C(y) = \frac{|C|}{|G|} \text{Li}(y) + O_K \left(y e^{-c_K \sqrt{\log y}} \right) \quad (13)$$

for some $c_K > 0$. The finitely many classes permit one choice of constants. Any exceptional term has the form $O_K(\text{Li}(y^{\beta_0}))$ with fixed $\beta_0 < 1$; it is absorbed in (13), since $y^{\beta_0-1} e^{c_K \sqrt{\log y}}$ is bounded for fixed K . Any lower threshold on y in the cited effective theorem is absorbed by enlarging the fixed-field constant. If $M = \mathbb{Q}$, use the ordinary prime number theorem in the same form.

Partial summation with the weight $\log^2 v$ gives

$$\begin{aligned} \sum_{\substack{p \leq y \\ \text{Frob}_p \in C}} \log^2 p &= \pi_C(y) \log^2 y - 2 \int_2^y \pi_C(v) \frac{\log v}{v} dv \\ &= \frac{|C|}{|G|} \int_2^y \log v dv + O_K(y) \\ &= \frac{|C|}{|G|} y \log y + O_K(y). \end{aligned}$$

The lower-endpoint constant is harmless. For the error, the boundary term is $O_K(y \log^2 y e^{-c_K \sqrt{\log y}}) = O_K(y)$, and the integral is bounded by a constant times $\int_2^y \log v e^{-c_K \sqrt{\log v}} dv = O_K(y)$. Multiply by the class value $P(C)^2$, sum over C , and use (12). This evaluates the first powers of unramified primes with leading coefficient r .

For each of the finitely many primes ramified in M ,

$$\sum_{p^j \leq y} A_K(p^j)^2 \leq n^2 (\log p)^2 \left\lfloor \frac{\log y}{\log p} \right\rfloor = O_K(\log y).$$

The total contribution of higher prime powers is bounded by

$$n^2 \sum_{2 \leq j \leq \log_2 y} \sum_{p \leq y^{1/j}} (\log p)^2 \ll_K y^{1/2} \log^3(2y) = O_K(y).$$

These estimates prove (10). □

3.3 An exact resolvent formula

Define the fixed constant

$$q_K = \frac{1}{2} \log d_K - \frac{r_1}{2} \log \pi - r_2 \log(2\pi), \quad \psi = \Gamma'/\Gamma.$$

Proposition 6 (Resolvent and uniform residual). *For $x \geq 1$ and $t \in \mathbb{R}$, put*

$$H_x(t) = 2 \sum_{\rho} \frac{x^{\rho-1/2-it}}{1 - (\rho - 1/2 - it)^2}, \quad P_x(t) = \sum_{m \geq 2} \frac{A_K(m)}{m^{1/2+it}} \min(m/x, x/m), \quad (14)$$

$$G_K(t) = 2q_K + \frac{r_1}{2} \{\psi(-1/4 + it/2) + \psi(3/4 - it/2)\} \\ + r_2 \{\psi(-1/2 + it) + \psi(3/2 - it)\} + Z_K(3/2 - it), \quad (15)$$

$$B_x(t) = x^{-1} G_K(t).$$

All nontrivial zeros, including real zeros, occur in H_x . There is an exact identity

$$H_x(t) = -P_x(t) + B_x(t) + R_x(t), \quad (16)$$

where, with $\mu_0 = r_1 + r_2 - 1$ and $\mu_j = r_2 + r_1 \mathbf{1}_{2|j}$ for $j \geq 1$,

$$R_x(t) = \frac{2x^{1/2-it}}{(1/2 + it)(3/2 - it)} + 2 \sum_{j \geq 0} \frac{\mu_j x^{-j-1/2-it}}{(j - 1/2 + it)(j + 3/2 + it)}. \quad (17)$$

The zero, prime, and trivial-zero series converge absolutely and locally uniformly on $[1, \infty) \times \mathbb{R}$. Uniformly on this domain,

$$G_K(t) = n \log(|t| + 2) + O_K(1), \quad (18)$$

$$R_x(t) \ll_K \frac{x^{1/2}}{1 + t^2} + \frac{x^{-1/2}}{|t| + 2}, \quad \int_{\mathbb{R}} |R_x(t)|^2 dt \ll_K x. \quad (19)$$

Proof. For $d > 1$ and $y > 0$, contour shifting gives

$$\frac{1}{2\pi i} \int_{(d)} \frac{2y^w}{1 - w^2} dw = \begin{cases} y^{-1} - y, & y > 1, \\ 0, & 0 < y \leq 1. \end{cases}$$

Indeed, for $y \geq 1$ one shifts to the left and picks up the residues $-y$ and y^{-1} at $w = 1$ and $w = -1$; for $y < 1$ one shifts to the right and crosses no pole. The quadratic denominator makes the horizontal integrals vanish, and the new vertical integral tends to zero. This also gives the stated value at $y = 1$. Apply this identity to

$$I_x(t) = \frac{1}{2\pi i} \int_{(2)} -Z_K(z) \frac{2x^{z-1/2-it}}{1 - (z - 1/2 - it)^2} dz.$$

Absolute convergence of the Dirichlet series yields

$$I_x(t) = \sum_{m < x} A_K(m) m^{-1/2-it} (m/x - x/m).$$

Fix an integer $\nu \geq 1$ and shift the contour to $\operatorname{Re} z = -\nu - 1/2$. We record the estimates that justify this shift. The order-one Hadamard product gives $\#\{\rho : |\rho| \leq R\} = O_K(R^{5/4})$. In every sufficiently large

interval $[Y, 2Y]$ one can therefore choose Y_1 whose distance from every zero ordinate with $|\rho| \leq 5Y$ is $\gg_K Y^{-1/4}$. On the horizontal segments at heights $\pm Y_1$, the logarithmic derivative of the Hadamard product is $O_{K,\nu}(Y^{3/2})$; zeros with $|\rho| > 5Y$ contribute $O_K(Y \sum_{|\rho| > 5Y} |\rho|^{-2})$, and the remaining gamma and rational terms are $O_{K,\nu}(\log Y)$. Since the Mellin kernel is $O_{x,t}(Y^{-2})$ there, both horizontal integrals tend to zero.

For the new vertical line, logarithmic differentiation of the functional equation gives

$$-Z_K(s) = Z_K(1-s) + 2q_K + \frac{r_1}{2} \{\psi(s/2) + \psi((1-s)/2)\} + r_2 \{\psi(s) + \psi(1-s)\}.$$

The logarithmic derivatives of the factors $s(s-1)$ cancel in this identity. On $s = -\nu - 1/2 + iu$, the reflection formula moves the negative-real-part gamma arguments to right half-planes; the resulting cotangents stay uniformly bounded because their real parts remain a fixed positive distance from the integers. Stirling's estimate therefore shows $Z_K(-\nu - 1/2 + iu) \ll_K \log(\nu + |u| + 2)$. Since the denominator on that line is at least $\nu^2 + (u-t)^2$, the left integral is

$$\ll_K x^{-\nu-1} \frac{\log(\nu + |t| + 2)}{\nu},$$

and vanishes as $\nu \rightarrow \infty$.

The crossed poles are the nontrivial zeros, the pole at 1, the trivial zeros $-j$ with multiplicities μ_j , and the two kernel poles $3/2 + it$ and $-1/2 + it$. For $j \geq 1$ the functional equation and the Euler product give $\xi_K(-j) = \xi_K(1+j) \neq 0$, so cancellation of the gamma poles gives $\mu_j = r_2 + r_1 \mathbf{1}_{2|j}$. At $j = 0$, the factor $s(s-1)$ reduces the order by one, giving $\mu_0 = r_1 + r_2 - 1$. Summing the residues gives

$$I_x(t) = -H_x(t) + R_x(t) + xZ_K(3/2 + it) - x^{-1}Z_K(-1/2 + it).$$

Comparing Dirichlet-series coefficients gives $-I_x(t) + xZ_K(3/2 + it) = -P_x(t)$, while the functional equation at $-1/2 + it$ gives $-Z_K(-1/2 + it) = G_K(t)$. This proves (16). Absolute convergence follows from $\sum_\rho (1 + |\rho|^2)^{-1} < \infty$, $\sum_m A_K(m)m^{-3/2} < \infty$, and $\sum_{j \geq 0} (j+1)^{-2} < \infty$; the same majorants are locally uniform in (x, t) .

Finally, Stirling and absolute convergence of $Z_K(3/2 - it)$ give (18). The elementary denominator bounds

$$|(1/2 + it)(3/2 - it)| \gg 1 + t^2, \quad |(j - 1/2 + it)(j + 3/2 + it)| \gg (j + 1)^2 + t^2$$

imply (19); squaring and integrating gives its L^2 estimate. $\square$

3.4 Plancherel and the sharp height window

Proposition 7 (Exact kernel and localization). *For $T \geq 3$ and $x > 0$, define*

$$A_T = \{\rho : 0 < \gamma \leq T\}, \quad F_K(x, T) = \sum_{\rho, \rho' \in A_T} x^{\rho - \rho'} W(\rho - \rho').$$

Here A_T is a multiset, so $|A_T| = N_K(T)$. For every $x > 0$, define the finite sum

$$H_{x,T}(t) = 2 \sum_{\rho \in A_T} \frac{x^{\rho - 1/2 - it}}{1 - (\rho - 1/2 - it)^2}.$$

Then

$$F_K(x, T) = \frac{1}{2\pi} \int_{\mathbb{R}} |H_{x,T}(t)|^2 dt \geq 0. \quad (20)$$

Uniformly for $x \geq 1$ and $T \geq 3$,

$$F_K(x, T) = \frac{1}{2\pi} \int_0^T |H_x(t)|^2 dt + O_K(x \log^3(T+2)). \quad (21)$$

The same local estimates also give the normalization

$$N_K(T) = \frac{n}{2\pi} T \log T + O_K(T). \quad (22)$$

No condition that T avoid a zero ordinate is imposed.

Proof. For $\delta \in (-1/2, 1/2)$ and $\gamma \in \mathbb{R}$ put $k_{\delta, \gamma}(t) = [1 + (t - \gamma + i\delta)^2]^{-1}$. Contour shifting gives

$$\widehat{k}_{\delta, \gamma}(v) = \pi e^{-2\pi|v|} e^{-2\pi\delta v} e^{-2\pi i \gamma v}.$$

With $\delta_\rho = \beta - 1/2$, Plancherel applied to

$$e^{it \log x} H_{x, T}(t) = 2 \sum_{\rho \in A_T} x^{\rho-1/2} k_{\delta_\rho, \gamma}(t)$$

and the involution $\rho' \mapsto 1 - \bar{\rho}'$ give

$$\int_{\mathbb{R}} |H_{x, T}(t)|^2 dt = 2\pi \sum_{\rho, \rho' \in A_T} x^{\rho-\rho'} W(\rho - \rho').$$

Here we used $\int_{\mathbb{R}} e^{-4\pi|v|} e^{-2\pi a v} dv = W(a)/(2\pi)$ for $|\operatorname{Re} a| < 1$. This proves (20).

Let $\mathcal{B}_T$ be the complementary multiset and set

$$S_A(t) = \sum_{\rho \in A_T} \frac{1}{1 + (t - \gamma)^2}, \quad S_B(t) = \sum_{\rho \in \mathcal{B}_T} \frac{1}{1 + (t - \gamma)^2}, \quad \ell = \log(T+2).$$

Unit-interval summation with Lemma 4 gives

$$\begin{aligned} S_A(t) &\ll_K \ell & (0 \leq t \leq T), \\ S_B(t) &\ll_K \ell \{(1+t)^{-1} + (1+T-t)^{-1}\} & (0 \leq t \leq T), \\ S_A(t) &\ll_K \ell \{1 + \operatorname{dist}(t, [0, T])\}^{-1} & (t \notin [0, T]). \end{aligned}$$

For example, the zeros above T are grouped into $(T+j, T+j+1]$, and those below the window into $[-j-1, -j)$. The resulting series are bounded using

$$\sum_{j \geq 0} \frac{1}{1 + (a+j)^2} \ll \frac{1}{1+a}, \quad \sum_{j > J} \frac{\log(j+3)}{j^2} \ll \frac{\log(J+3)}{J}.$$

The second estimate handles ordinates larger than a fixed multiple of T ; up to that point every local count is $O_K(\ell)$. The estimate outside $[0, T]$ follows in the same way by grouping the ordinates in A_T into unit intervals measured from the nearer endpoint. Since $|1 + (y + i\delta)^2| \geq \frac{3}{4}(1 + y^2)$, these bounds imply

$$\int_0^T S_A S_B \ll_K \ell^3, \quad \int_0^T S_B^2 \ll_K \ell^2, \quad \int_{\mathbb{R} \setminus [0, T]} S_A^2 \ll_K \ell^2.$$

Indeed, $|H_{x, T}| \leq (8/3)\sqrt{x} S_A$ and $|H_x - H_{x, T}| \leq (8/3)\sqrt{x} S_B$ for $x \geq 1$. Expanding the two squared norms and applying the three displayed integral estimates proves (21).

Finally, for $h_\rho(t) = (2 - \beta)/((2 - \beta)^2 + (t - \gamma)^2)$ one has $\int_{\mathbb{R}} h_\rho = \pi$. The same endpoint sums show

$$\int_0^T U_K(t) dt = \pi N_K(T) + O_K(\ell^2).$$

More explicitly, for $0 < \gamma \leq T$ the missing mass is $O((1 + \gamma)^{-1} + (1 + T - \gamma)^{-1})$; its sum is $O_K(\ell^2)$. The contribution on $[0, T]$ of all excluded zeros is at most $2 \int_0^T S_B(t) dt = O_K(\ell^2)$. All terms are nonnegative, so Tonelli's theorem justifies the interchanges. Integrating (7) proves (22). $\square$

3.5 The three mean-square estimates

For the remainder of this section, $\langle U, V \rangle_2 = \int_0^T U(t) \overline{V(t)} dt$ and $\|U\|_2^2 = \langle U, U \rangle_2$.

Proposition 8 (Mean-square inputs). *Uniformly for $x \geq 1$ and $T \geq 3$, writing $L = \log T$,*

$$\|P_x\| * 2^2 = Tr \log x + O_K(T + x \log(2x)), \quad (23)$$

$$\|B_x\| 2^2 = n^2 T x^{-2} L^2 + O_K(T x^{-2} L), \quad (24)$$

$$\|R_x\| 2^2 \ll_K x. \quad (25)$$

In particular, for $1 \leq x \leq T$ and $u = \log(2x)$,

$$\begin{aligned} |\langle P_x, B_x \rangle_2| &\ll_K TL \sqrt{u}/x, \\ |\langle P_x, R_x \rangle_2| &\ll_K \sqrt{Txu}, \\ |\langle B_x, R_x \rangle_2| &\ll_K \sqrt{T/x} L. \end{aligned} \quad (26)$$

Proof. Put

$$a_x(m) = A_K(m) m^{-1/2} \min(m/x, x/m), \quad D(x) = \sum_m a_x(m)^2, \quad V(x) = \sum_m m a_x(m)^2.$$

Stieltjes partial summation in (10), separately over $m \leq x$ and $m > x$, gives

$$\begin{aligned} D(x) &= x^{-2} \sum_{m \leq x} m A_K(m)^2 + x^2 \sum_{m > x} \frac{A_K(m)^2}{m^3} = r \log x + O_K(1), \\ V(x) &= x^{-2} \sum_{m \leq x} m^2 A_K(m)^2 + x^2 \sum_{m > x} \frac{A_K(m)^2}{m^2} \ll_K x \log(2x). \end{aligned}$$

For the first identity, the two ranges each contribute $(r/2) \log x + O_K(1)$. For the second, the first range is at most $B_K(x)$ and partial summation gives $\sum_{m > x} A_K(m)^2 m^{-2} \ll_K \log(2x)/x$. Thus

$$\sum_m m |a_x(m)|^2 < \infty.$$

The Montgomery–Vaughan mean-value theorem [13, Corollary 3], first for finite sums and then by L^2 convergence of the tails, therefore proves (23). From (18), $G_K(t) = n \log(t + 2) + O_K(1)$ for $t \geq 0$, whence

$$\int_0^T |G_K(t)|^2 dt = n^2 T L^2 + O_K(TL).$$

This proves (24), while (25) follows from (19). For $1 \leq x \leq T$ the resulting norm bounds are

$$\|P_x\|_2 \ll_K \sqrt{T \log(2x)}, \quad \|B_x\|_2 \ll_K \sqrt{T} L/x, \quad \|R_x\|_2 \ll_K \sqrt{x}.$$

Cauchy–Schwarz gives (26). $\square$

3.6 A uniform form-factor estimate

Proposition 9 (Complete error budget). *Let $T \geq 3$, $L = \log T$, and $u = \log(2x)$. Uniformly for $1 \leq x \leq T$,*

$$2\pi F_K(x, T) = n^2 T x^{-2} L^2 + r T \log x + O_K(\mathcal{E}(x, T)), \quad (27)$$

$$\begin{aligned} \mathcal{E}(x, T) = & T x^{-2} L + T + x u + T L \sqrt{u}/x \\ & + \sqrt{T x u} + \sqrt{T/x} L + x + x L^3. \end{aligned} \quad (28)$$

Proof. Expand the exact identity (16) in $L^2(0, T)$:

$$\begin{aligned} \|H_x\|_2^2 = & \|P_x\|_2^2 + \|B_x\|_2^2 + \|R_x\|_2^2 - 2 \operatorname{Re} \langle P_x, B_x \rangle_2 \\ & - 2 \operatorname{Re} \langle P_x, R_x \rangle_2 + 2 \operatorname{Re} \langle B_x, R_x \rangle_2. \end{aligned}$$

The two main terms come from (23) and (24). Their errors give $T + x u + T x^{-2} L$. The residual norm gives x , and (26) gives the three square-root terms. Finally, (21) contributes $x L^3$, since $\log(T+2) \asymp L$ for $T \geq 3$. These are exactly all the terms in (28). $\square$

3.7 Quantitative Fourier synthesis

Proposition 10 (Integration at fixed strict support). *For $0 < b < 1$ and f as in Lemma 3, let $S_{K,f}(T)$ denote the pair sum on the left of (6), and put $Q_K(T) = n T \log T / (2\pi)$. Then*

$$\frac{S_{K,f}(T)}{Q_K(T)} = n f(0) + \frac{r}{n} \int_{-1}^1 |a| f(a) da + O_{K,b,f}(1/\log T). \quad (29)$$

Replacing $Q_K(T)$ by $N_K(T)$ gives (6) and hence proves Lemma 3.

Proof. The zero sum over A_T is finite. Therefore the Fourier convention and $x^z = \exp(z \log x)$ give exactly

$$S_{K,f}(T) = \int_{-b}^b f(a) F_K(T^a, T) da.$$

Exchanging the two zero variables, and using $W(-z) = W(z)$, gives $F_K(x^{-1}, T) = F_K(x, T)$. Since f is even,

$$S_{K,f}(T) = 2 \int_0^b f(a) F_K(T^a, T) da. \quad (30)$$

This identity applies to the complex Fourier arguments without any assumption on β beyond the critical strip.

Now insert (27) with $x = T^a = e^{aL}$ and divide by $Q_K(T)$. The contribution of the first main term is

$$2nL \int_0^b f(a) e^{-2aL} da = n f(0) + O_{K,f}(L^{-1}).$$

For detail, extend f by zero as already prescribed outside $[-b, b]$, set $v = 2La$, and use $|f(v/(2L)) - f(0)| \leq \|f'\|_\infty v/(2L)$ in the integral against $e^{-v} dv$ on $[0, \infty)$. This also includes its tail beyond $2bL$. The second main term contributes exactly

$$\frac{2r}{n} \int_0^b a f(a) da = \frac{r}{n} \int_{-1}^1 |a| f(a) da.$$

For the error terms, use

$$\int_0^b e^{-aL} \sqrt{\log 2 + aL} da = \frac{1}{L} \int_0^{bL} e^{-v} \sqrt{\log 2 + v} dv \ll L^{-1},$$

along with $\int_0^b e^{-2aL} da \leq 1/(2L)$, $\log 2 + aL \ll_b L$. In the order in which the eight terms occur in (28), their integrals after division by TL are

$$O_b \left(L^{-1}, L^{-1}, T^{b-1}, L^{-1}, T^{(b-1)/2} L^{-1/2}, T^{-1/2}, T^{b-1} L^{-1}, T^{b-1} L^2 \right).$$

For fixed $b < 1$, every entry is $O_b(L^{-1})$. This proves (29). Finally (22) gives, for sufficiently large T depending on K ,

$$\frac{Q_K(T)}{N_K(T)} = 1 + O_K(L^{-1}).$$

Multiplying (29) by this ratio changes its error by $O_{K,b,f}(L^{-1})$, which is already included. This proves (6). $\square$

The field and test function are fixed before $T \rightarrow \infty$. In Section 4, each endpoint cutoff is therefore held fixed while the limit is taken; no uniformity as $b \uparrow 1$ is required.

4 Weight removal and the exact scalar optimum

We first convert the weighted pair formula into the energy required by Lemma 2, then minimize its limiting scalar functional. Finally, fixed smooth cutoffs transfer that minimum to the zero counts. Throughout, K is fixed, $n \geq 1$ and $1 \leq r \leq n$ have the meanings in Section 1.1, $T \geq 3$, $L = \log T$, and $I = [-1/2, 1/2]$.

4.1 Exact weight removal and the limiting energy

Let $\eta \in C_c^\infty((-1/2, 1/2))$ be real and even, normalized by $\int_I \eta(x)^2 dx = 1$. Extend it by zero to $\mathbb{R}$, and set

$$p = \eta^2, \quad q = p * p, \quad (p * p)(u) = \int_{\mathbb{R}} p(v)p(u-v) dv.$$

Choose a with $0 < a < 1/2$ and $\text{supp } p \subset [-a, a]$. Thus q and q'' are fixed, real, even C_c^∞ functions supported in $[-2a, 2a] \subset (-1, 1)$; both satisfy Lemma 3. With $z = i(\rho - \rho')L/(2\pi)$, Fourier differentiation gives exactly

$$\left(\widehat{q}(z) - \frac{\widehat{q''}(z)}{4L^2} \right) W(\rho - \rho') = \widehat{p}(z)^2. \quad (31)$$

Indeed, compact support gives $\widehat{q} = (\widehat{p})^2$ at every complex argument. Integration by parts twice, with no boundary terms, gives $\widehat{q''}(z) = -4\pi^2 z^2 \widehat{q}(z)$, and $\pi^2 z^2 / L^2 = -(\rho - \rho')^2 / 4$. The multiplier on the left therefore cancels W exactly, also on the diagonal $\rho = \rho'$. This is Lamzouri's differential weight-removal device [11, Section 3, proof of Lemma 3.2].

Using the distinct zero set and multiplicities of Section 1.1, define the unweighted energy by

$$E_K(\eta, T) = \sum_{\rho, \rho' \in \mathcal{Z}_K(T)} m_\rho m_{\rho'} \widehat{p} \left(i(\rho - \rho') \frac{L}{2\pi} \right)^2.$$

The square is an ordinary square, not an absolute square. The sum is over ordered pairs, including the diagonal. If $S_{K,g}(T)$ denotes the weighted pair sum from Section 3, then (31) yields the finite-sum identity

$$E_K(\eta, T) = S_{K,q}(T) - \frac{S_{K,q''}(T)}{4L^2}.$$

Apply Lemma 3 separately to the two fixed tests. In particular $S_{K,q''}(T)/N_K(T) = O_{K,\eta}(1)$, so its contribution above is $O_{K,\eta}(L^{-2})$. No uniformity in a moving test is required. Evenness of p gives

$$q(0) = \int_I p(x)^2 dx, \quad \int_{-1}^1 |u|q(u) du = \iint_{I^2} |x+y|p(x)p(y) dx dy = \iint_{I^2} |x-y|p(x)p(y) dx dy.$$

The first double integral follows from Fubini and the second from $y \mapsto -y$. Consequently, for each fixed η ,

$$\frac{E_K(\eta, T)}{N_K(T)} = J_{n,r}(p) + o(1), \quad J_{n,r}(p) = n \int_I p^2 + \frac{r}{n} \iint_{I^2} |x-y|p(x)p(y) dx dy,$$

as $T \rightarrow \infty$. In fact Section 3 gives an $O_{K,\eta}(L^{-1})$ error here. Positivity of the total energy will follow from its identification with Lemma 2 below; no pointwise positivity of the complex summands is being used.

4.2 The scalar minimizer and a strict variational certificate

For this variational step, extend $J_{n,r}$ to the real affine space

$$\mathcal{A} = \{p \in L^2(I; \mathbb{R}) : \int_I p(x) dx = 1\}.$$

The double integral is absolutely convergent because $|x-y| \leq 1$ on I^2 and $L^2(I) \subset L^1(I)$. Members of $\mathcal{A}$ need not be even or nonnegative; this enlargement is only for the minimization, not an assertion that the pair formula applies to every member. Put

$$\omega = \frac{\sqrt{2r}}{n}, \quad p_*(x) = \frac{\omega \cos(\omega x)}{2 \sin(\omega/2)}, \quad x \in I.$$

Since $0 < \omega/2 \leq 1/\sqrt{2} < \pi/2$, the density p_* is positive on the closed interval and is smooth and even there. Direct integration gives $\int_I p_* = 1$. Define the potential of $p \in \mathcal{A}$ by

$$\Phi_p(x) = np(x) + \frac{r}{n} \int_I |x-y|p(y) dy, \quad \text{so that} \quad J_{n,r}(p) = \int_I \Phi_p(x)p(x) dx.$$

The identity $\partial_x^2|x-y| = 2\delta_y$, or differentiation of the two integrals on either side of $y = x$, gives $\Phi_{p_*}'' = np_*'' + (2r/n)p_* = 0$ in the interior of I , because $p_*'' = -\omega^2 p_*$ and $n\omega^2 = 2r/n$. Thus Φ_{p_*} is affine, and evenness makes it constant. Continuity extends that constant to the endpoints. Since $\int_I yp_*(y) dy = 0$, evaluation at $x = 1/2$ gives

$$\Phi_{p_*}(1/2) = np_*(1/2) + \frac{r}{2n} = \frac{n\omega}{2} \cot \frac{\omega}{2} + \frac{r}{2n} = \frac{r}{2n} + \sqrt{\frac{r}{2}} \cot \left(\frac{1}{n} \sqrt{\frac{r}{2}} \right) = C_{n,r}.$$

Multiplication by p_* and integration show $J_{n,r}(p_*) = C_{n,r}$.

Let $h \in L^2(I; \mathbb{R})$ with $\int_I h = 0$, and set

$$H(x) = \int_{-1/2}^x h(y) dy, \quad U_h(x) = \int_I |x-y|h(y) dy.$$

Then $H \in H_0^1(I)$: it is absolutely continuous, has weak derivative $h \in L^2$, and vanishes at both endpoints. Differentiating U_h gives $U'_h(x) = 2H(x)$ almost everywhere. Hence integration by parts, whose boundary term vanishes because H does, gives

$$\iint_{I^2} |x - y| h(x) h(y) dx dy = \int_I H'(x) U_h(x) dx = -2 \int_I H(x)^2 dx.$$

The linear term in $J_{n,r}(p_* + h)$ is $2 \int_I \Phi_{p_*}(x) h(x) dx = 2C_{n,r} \int_I h = 0$. The Dirichlet Poincaré inequality on the interval of length one, $\int_I H^2 \leq \pi^{-2} \int_I |H'|^2$, therefore yields

$$J_{n,r}(p_* + h) - C_{n,r} = n \int_I h^2 - \frac{2r}{n} \int_I H^2 \geq \left(n - \frac{2r}{n\pi^2} \right) \int_I h^2. \quad (32)$$

Since $r \leq n$, the coefficient is at least $n - 2/\pi^2 > 0$. Thus p_* is the unique minimizer on $\mathcal{A}$, up to equality almost everywhere, even among signed, not necessarily even functions.

This is the generalized Montgomery–Taylor constant of [8, Theorem 5] with $c_1 = n$, $c_2 = r/n$, and $\Delta = 1$. The direct argument is retained because it also supplies the minimizer and the strictness needed below.

For later use, Cauchy–Schwarz gives $\int_I p_*^2 \geq 1$, and the distance integral of the positive density p_* is strictly positive. Thus $C_{n,r} > n$. The uniform density $p = 1$ on I satisfies

$$\iint_{I^2} |x - y| dx dy = 2 \int_{-1/2}^{1/2} \int_{-1/2}^x (x - y) dy dx = \frac{1}{3}, \quad J_{n,r}(1) = n + \frac{r}{3n}.$$

It differs from p_* because $\omega > 0$, so (32) gives

$$n < C_{n,r} < n + \frac{r}{3n} \leq n + \frac{1}{3}.$$

In particular $\lfloor C_{n,r} \rfloor = n$. By the threshold comparison in Section 2, $k = n$ is the unique maximizing integer for the bound $(2k + 1 - C_{n,r})/[k(k + 1)]$.

4.3 Smooth cutoffs and the distinct-zero consequence

Proof of Theorem 1. The zero extension of p_* is not a smooth compactly supported test, so we do not apply Lemma 3 to it or to its endpoint convolution. For integers $j \geq 5$, choose real even $\chi_j \in C_c^\infty((-1/2 + 1/j, 1/2 - 1/j))$ such that $0 \leq \chi_j \leq 1$ and $\chi_j = 1$ on $|x| \leq 1/2 - 2/j$. Define

$$c_j = \int_I \chi_j^2 p_*, \quad \eta_j = \frac{\chi_j \sqrt{p_*}}{\sqrt{c_j}}, \quad p_j = \eta_j^2 = \frac{\chi_j^2 p_*}{c_j},$$

with zero extensions outside I . Positivity and smoothness of p_* give $c_j > 0$ and real even $\eta_j \in C_c^\infty((-1/2, 1/2))$ with $\int_I \eta_j^2 = 1$. Dominated convergence gives $c_j \rightarrow 1$ and $p_j \rightarrow p_*$ in both $L^1(I)$ and $L^2(I)$.

For the distance form $\mathcal{D}(p) = \iint_{I^2} |x - y| p(x) p(y) dx dy$, the bound

$$|\mathcal{D}(p) - \mathcal{D}(v)| \leq \|p - v\| * L^1(I) (\|p\| * L^1(I) + \|v\|_{L^1(I)})$$

follows by expanding the difference and using $|x - y| \leq 1$. Together with L^2 convergence, it proves $C_j := J_{n,r}(p_j) \rightarrow J_{n,r}(p_*) = C_{n,r}$.

For each sufficiently large T , map the zero multiset into $\mathbb{C}$ by

$$z_\rho = i(\rho - 1/2) \frac{L}{2\pi}.$$

This map is injective, preserves multiplicities, and satisfies $z_{1-\bar{\rho}} = \overline{z_\rho}$. Reflection preserves the window $0 < \gamma \leq T$, so the transformed multiset satisfies the hypotheses of Lemma 2. Its N and D are exactly $N_K(T)$ and $D_K(T)$, and its energy, with $K_{\eta_j} = \hat{p}_j$, is exactly $E_K(\eta_j, T)$. For the fixed integer $k = n$, that lemma gives

$$\frac{D_K(T)}{N_K(T)} \geq \frac{2n + 1 - E_K(\eta_j, T)/N_K(T)}{n(n + 1)}.$$

For every fixed j , the energy limit proved above therefore yields

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq \frac{2n + 1 - C_j}{n(n + 1)}.$$

Only now let $j \rightarrow \infty$. The left side is independent of j , so this proves (1). No estimate uniform in the cutoff index is used, and $k = n$ need not be optimal for each individual C_j . Taking a cutoff family of the uniform density in the same way gives

$$\frac{2n + 1 - (n + r/(3n))}{n(n + 1)} = \frac{1}{n} - \frac{r}{3n^2(n + 1)} \geq \frac{3n + 2}{3n(n + 1)},$$

which proves (2). The strict comparison with $J_{n,r}(1)$ proves the asserted strict improvement in Theorem 1; its geometric increment is proved separately in Section 5.

Finally, the dependence on rank can be checked without differentiating the cotangent. For fixed n and $1 \leq r' < r'' \leq n$, let $p_{r''}$ be the positive minimizer for $J_{n,r''}$. Then

$$C_{n,r''} = J_{n,r'}(p_{r''}) + \frac{r'' - r'}{n} \mathcal{D}(p_{r''}) > C_{n,r'},$$

because $\mathcal{D}(p_{r''}) > 0$ and $J_{n,r'}(p_{r''}) \geq C_{n,r'}$. Hence $b_{n,r}$ decreases strictly with r , and the specialization $r = n$ gives the degree-only bounds in Table 1. These are bounds for distinct complex zero locations; this argument does not replace D_K by a count of simple or critical-line zeros. $\square$

5 A strict geometric refinement without GRH

Throughout this section, K is a fixed number field of degree $n \geq 2$, and $r = |H \backslash G / H|$ has the meaning fixed in Section 1.1; in particular $1 \leq r \leq n$. The counts $D_K(T)$ and $N_K(T)$ concern distinct complex zero locations and zeros with multiplicity, respectively. The refinement below is therefore a distinct-zero bound, not a simple-zero or critical-line proportion. Its additional input is the bounded interval occupied by the real elements of the normalized zero multiset.

Theorem 11. *If $n \geq 2$, there is an explicit constant $\epsilon_{n,r} > 0$ such that*

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq \frac{2n + 1 - C_{n,r}}{n(n + 1)} + \epsilon_{n,r}. \quad (33)$$

Proof. Put $C = C_{n,r}$, $b_0 = (2n + 1 - C)/(n(n + 1))$, and

$$\begin{aligned}\lambda &= \frac{\sqrt{r/2}}{n}, & c &= \frac{\lambda \tan \lambda}{\pi}, \\ \delta &= \min \left\{ \left(\frac{\lambda^2}{\pi^2 - \lambda^2} \right)^2, \frac{6\lambda^4}{81\pi^4(1 + c^2)^3} \right\}, & q_0 &= \frac{b_0}{3} - \frac{2}{9n}, \\ a &= 2\delta q_0, & B &= (2n + 6)\sqrt{C}.\end{aligned}$$

We will show that $q_0 > 0$ and that one may take

$$\epsilon_{n,r} = \frac{1}{n(n+1)} \left(\frac{2a}{\sqrt{B^2 + 4a} + B} \right)^2.$$

Step 1: an explicit three-point kernel bound. Write

$$p_\lambda(u) = \frac{\lambda}{\sin \lambda} \cos(2\lambda u) \mathbf{1}_{[-1/2, 1/2]}(u), \quad K_\lambda = \widehat{p}_\lambda.$$

Since $1 \leq r \leq n$ and $n \geq 2$, we have $0 < \lambda \leq 1/2$. Thus p_λ is nonnegative, even, and has integral one; in particular K_λ is real on $\mathbb{R}$ and $K_\lambda(0) = 1$. For any $R > 2$, set

$$\delta_R = \min \left\{ \left(\frac{\lambda^2}{\pi^2 - \lambda^2} \right)^2, \frac{6\lambda^4}{\pi^4 R^4 (1 + c^2)^3} \right\}.$$

This constant is positive and less than one. Order any three real locations of diameter at most R and let u, v be their consecutive gaps. Then

$$K_\lambda(u)^2 + K_\lambda(v)^2 + K_\lambda(u+v)^2 \geq \delta_R, \quad u, v \geq 0, \quad u+v \leq R. \quad (34)$$

Direct integration gives the nonsingular formula

$$K_\lambda(t) = \frac{\lambda}{2 \sin \lambda} \left\{ \frac{\sin(\lambda - \pi t)}{\lambda - \pi t} + \frac{\sin(\lambda + \pi t)}{\lambda + \pi t} \right\},$$

where $\sin v/v$ is assigned the value one at $v = 0$. Equivalently,

$$K_\lambda(t) = \frac{\lambda^2 \cos(\pi t) - \pi t \lambda \cot \lambda \sin(\pi t)}{\lambda^2 - \pi^2 t^2},$$

with removable singularities at $t = \pm \lambda/\pi$. The integral representation gives, for $0 < t < 1$,

$$K'_\lambda(t) = -4\pi \int_0^{1/2} u p_\lambda(u) \sin(2\pi t u) du < 0.$$

Hence K_λ decreases on $[0, 1]$ and $K_\lambda(1) = \lambda^2/(\pi^2 - \lambda^2) > 0$. If $u \leq 1$ or $v \leq 1$, the first term in δ_R suffices. Otherwise $u, v > 1$. Put $s = u + v$, $A = u^2 + uv + v^2$, and $z_t = e^{i\pi t}(t - ic)$. The exponential phases cancel, and multiplication gives the exact identity

$$z_u z_v \bar{z}_s = (u - ic)(v - ic)(s + ic) = uvs - ic(A + c^2).$$

For unit complex numbers α, β , $|\operatorname{Im}(\alpha\beta)| \leq |\operatorname{Im} \alpha| + |\operatorname{Im} \beta|$; iterating gives the corresponding three-factor inequality. Furthermore, for $t > 1$,

$$\frac{|\operatorname{Im} z_t|}{|z_t|} = \frac{\pi^2 t^2 - \lambda^2}{\pi \lambda \cot \lambda \sqrt{t^2 + c^2}} |K_\lambda(t)| \leq \frac{\pi t}{\lambda \cot \lambda} |K_\lambda(t)|, \quad |z_t| \leq t \sqrt{1 + c^2}.$$

Apply the three-factor inequality to $z_u/|z_u|$, $z_v/|z_v|$, and $\bar{z}_s/|z_s|$. Cauchy–Schwarz and $u^2 + v^2 + s^2 = 2A$ then give

$$\frac{c(A + c^2)}{uvs(1 + c^2)^{3/2}} \leq \frac{\pi}{\lambda \cot \lambda} (u|K_\lambda(u)| + v|K_\lambda(v)| + s|K_\lambda(s)|) \leq \frac{\pi\sqrt{2A}}{\lambda \cot \lambda} \left(\sum_{t=u,v,s} K_\lambda(t)^2 \right)^{1/2}.$$

Squaring, using $c\lambda \cot \lambda = \lambda^2/\pi$ and $(A + c^2)/A \geq A$, yields

$$\sum_{t=u,v,s} K_\lambda(t)^2 \geq \frac{\lambda^4}{2\pi^4(1 + c^2)^3} \frac{u^2 + uv + v^2}{u^2v^2(u + v)^2} \geq \frac{6\lambda^4}{\pi^4 R^4(1 + c^2)^3}.$$

For the last inequality put $w = uv \leq s^2/4$; the numerator $s^2 - w$ is at least $3s^2/4$, while $w^2s^2 \leq s^6/16$. Hence $(s^2 - w)/(w^2s^2) \geq 12/s^4 \geq 12/R^4$. This proves (34) without numerical minimization.

Step 2: projection away from the nonreal vectors. We now make a finite argument for an arbitrary real, even, compactly supported $\eta \in L^2(\mathbb{R})$ with $\int \eta^2 = 1$. Use K_η and a nonempty finite conjugation-invariant multiset Z with support Ω and positive integer multiplicities m_z , where $m_{\bar{z}} = m_z$. Retain N, D, N_C and the real Hilbert space $\mathcal{H}_0$ from Lemma 2. The kernel in this step need not be K_λ ; we will apply the argument to smooth approximations in Step 5. With the functions of Lamzouri [11, proof of Proposition 2.1], the real symmetric finite-rank operator from that lemma decomposes as

$$F = F_R + F_C, \quad F_R = \sum_{x \in \Omega \cap \mathbb{R}} m_x f_x \otimes f_x, \quad F_C = 2 \sum_{z \in \mathcal{P}} m_z (g_z \otimes g_z - h_z \otimes h_z),$$

where $\mathcal{P}$ contains one representative of each nonreal conjugate pair, and $E = \|F\|_{\text{HS}}^2$. Let N_C be the nonreal mass, $c_1 = |\mathcal{P}|$, and D_R the number of distinct real locations. Thus $D = D_R + 2c_1$, $N_C = 2 \sum_{z \in \mathcal{P}} m_z$, and $\text{tr } F_C = N_C$, since $\|g_z\|^2 - \|h_z\|^2 = 1$. Let Π project orthogonally onto the real span of all g_z, h_z with $z \in \mathcal{P}$, and set $\Pi^\perp = I - \Pi$. Since F_C is symmetric, its range and the range of its adjoint (which is the same range) lie in this span, so $\Pi F_C = F_C \Pi = F_C$ and $\Pi^\perp F_C \Pi^\perp = 0$. The lost real mass $\mathcal{L} = \sum_{x \in \Omega \cap \mathbb{R}} m_x \|\Pi f_x\|^2$ satisfies

$$\mathcal{L} + N_C = \text{tr}(\Pi F), \quad \mathcal{L} \leq \sqrt{2c_1 E} - N_C \leq \sqrt{N_C E} - N_C. \quad (35)$$

Indeed, Hilbert–Schmidt Cauchy–Schwarz gives

$$0 \leq \text{tr}(\Pi F) \leq \|\Pi\| * \text{HS} \|F\| * \text{HS} = \sqrt{\text{rank } \Pi} \sqrt{E},$$

because $\text{tr}(\Pi F) = \mathcal{L} + N_C \geq 0$, and $\text{rank } \Pi \leq 2c_1 \leq N_C$. Compression by an orthogonal projection contracts the Hilbert–Schmidt norm, so

$$\|\Pi^\perp F_R \Pi^\perp\| * \text{HS}^2 = \|\Pi^\perp F \Pi^\perp\| * \text{HS}^2 \leq E.$$

Neither this contraction nor the trace estimate requires F to be positive semidefinite. If there are no nonreal locations, take $\Pi = 0$; the same identities hold with $N_C = c_1 = \mathcal{L} = 0$.

Step 3: retaining disjoint triples after compression. Fix an integer threshold $k \geq 1$. For every integer $m \geq 1$ and $0 \leq t \leq 1$,

$$m^2 t^2 \geq (2k + 1)m - k(k + 1) - 2(k + 1)m(1 - t).$$

After subtraction the difference is $(mt - k - 1)^2 + m - k - 1$. It is nonnegative when $m \geq k + 1$. When $m \leq k$, put $q = k + 1 - m \geq 1$; since $mt \leq m$, the difference is at least $q^2 - q \geq 0$. For a real location x , let $\ell_x = \|\Pi f_x\|^2 \in [0, 1]$. Then $\|\Pi^\perp f_x\|^2 = 1 - \ell_x$. For two real locations, write

$a_{xy} = \langle f_x, f_y \rangle = K_\eta(x - y)$ and $b_{xy} = \langle \Pi^\perp f_x, \Pi^\perp f_y \rangle$. These are real numbers of absolute value at most one, and $a_{xy} - b_{xy} = \langle \Pi f_x, \Pi f_y \rangle$. Consequently

$$|K_\eta(x - y)^2 - b_{xy}^2| = |a_{xy} - b_{xy}| |a_{xy} + b_{xy}| \leq 2\sqrt{\ell_x \ell_y} \leq \ell_x + \ell_y.$$

Suppose we select a set $\mathfrak{T}$ of vertex-disjoint triples of distinct real locations, each triple satisfying

$$\sum_{\{x,y\} \text{ in the triple}} K_\eta(x - y)^2 \geq d \quad \text{for some fixed } 0 < d < 1.$$

Let $\mathfrak{E}$ be the set of the $3|\mathfrak{T}|$ unordered edges of the selected triples. The compressed real energy expands as

$$\|\Pi^\perp F_R \Pi^\perp\| * \text{HS}^2 = \sum_x m_x^2 (1 - \ell_x)^2 + 2 \sum * \{x, y\} \subset \Omega \cap \mathbb{R} m_x m_y b_{xy}^2,$$

where the second sum runs over all two-element subsets. Apply the scalar inequality with $t = 1 - \ell_x$ to each diagonal term, retain only the selected edges in the second sum, discard their factors $m_x m_y \geq 1$ from the nonnegative squares, and only then apply the edge-loss estimate. The result is

$$\|\Pi^\perp F_R \Pi^\perp\| * \text{HS}^2 \geq (2k + 1)(N - N_C) - k(k + 1)D_R - 2(k + 1)\mathcal{L} + 2d|\mathfrak{T}| - 2 \sum * \{x, y\} \in \mathfrak{E} (\ell_x + \ell_y).$$

Each selected vertex has degree two in $\mathfrak{E}$, so $\sum_{\{x,y\} \in \mathfrak{E}} (\ell_x + \ell_y) = 2 \sum_{\text{selected } x} \ell_x \leq 2\mathcal{L}$. Thus the diagonal loss is $2(k + 1)\mathcal{L}$ and the ordered cross-term loss is $4\mathcal{L}$. We conclude that

$$E \geq (2k + 1)(N - N_C) - k(k + 1)D_R + 2d|\mathfrak{T}| - (2k + 6)\mathcal{L}. \quad (36)$$

Step 4: packing the height interval. Suppose that the real locations lie in $[\theta_0, \theta_0 + Y_0]$ and that the kernel K_η satisfies the preceding triple bound with constant d whenever the diameter is at most R . Assign a real location x to cell $\lfloor (x - \theta_0)/R \rfloor$. There are at most $\lfloor Y_0/R \rfloor + 1 \leq Y_0/R + 1$ cells, each of diameter less than R ; if Y_0/R is an integer, this includes the possible final singleton endpoint. Taking $\lfloor s/3 \rfloor$ triples in a cell containing s distinct locations and using $\lfloor s/3 \rfloor \geq s/3 - 2/3$ yields

$$|\mathfrak{T}| \geq \frac{D_R}{3} - \frac{2Y_0}{3R} - \frac{2}{3} \geq \frac{D}{3} - \frac{N_C}{3} - \frac{2Y_0}{3R} - \frac{2}{3},$$

where the last step uses $D_R = D - 2c_1 \geq D - N_C$. The nonreal defect (4) gives $\Delta_k := E - (2k + 1)N + k(k + 1)D \geq N_C$. Before discarding terms, (36) and $D = D_R + 2c_1$ give

$$\Delta_k \geq -(2k + 1)N_C + 2k(k + 1)c_1 + 2d|\mathfrak{T}| - (2k + 6)\mathcal{L}.$$

Substitute (35), use the packing bound, and discard the nonnegative $2k(k + 1)c_1$ term. This gives

$$\begin{aligned} \Delta_k &\geq 2d \left(\frac{D}{3} - \frac{2Y_0}{3R} - \frac{2}{3} \right) - (2k + 6)\sqrt{N_C E} \\ &\quad + \left(5 - \frac{2d}{3} \right) N_C. \end{aligned} \quad (37)$$

Here $5 = (2k + 6) - (2k + 1)$, and the subtraction $2d/3$ comes from the nonreal term in the packing bound. The coefficient $5 - 2d/3$ is positive. These inequalities also cover an empty real set, fewer than three real locations, and dependent spanning vectors: use $\mathfrak{T} = \emptyset$ when no triple exists, and note that dependence only lowers the rank of Π . A negative packing lower bound is harmless.

Step 5: fixed smooth kernels and the limiting bound. For the zero transformation

$$z_\rho = i(\rho - \tfrac{1}{2}) \frac{\log T}{2\pi},$$

the symmetry $\rho \mapsto 1 - \bar{\rho}$ becomes conjugation and preserves the positive-height window and multiplicities. Real elements correspond to $\operatorname{Re} \rho = 1/2$ and lie in $[-T \log T / (2\pi), 0)$. Thus we may take $\theta_0 = -T \log T / (2\pi)$ and $Y_0 = T \log T / (2\pi)$; the zero-counting asymptotic (22) gives $Y_0 / N_K(T) \rightarrow 1/n$. We take $k = n$ and $R = 3$, so $\delta_R = \delta$ for the endpoint kernel. First note that the limiting scalar parameters satisfy $q_0 > 0$: the optimized bound is strictly larger than (2), and $r \leq n$, so

$$q_0 = \frac{b_0}{3} - \frac{2}{9n} > \frac{3n+2}{9n(n+1)} - \frac{2}{9n} = \frac{1}{9(n+1)} > 0.$$

We apply the analytic formula only to the fixed smooth cutoffs $p_j = \eta_j^2 \rightarrow p_\lambda$ in $L^1 \cap L^2$ constructed in Section 4. Each η_j is real, even, smooth, supported in $(-1/2, 1/2)$, and normalized by $\int \eta_j^2 = 1$. Put $K_j = \hat{p}_j$ and $e_j = \|p_j - p_\lambda\|_1$. Then $C_j = J_{n,r}(p_j) \rightarrow C$, $e_j \rightarrow 0$, and $\sup_{t \in \mathbb{R}} |K_j(t) - K_\lambda(t)| \leq e_j$. For $u, v \geq 0$ with $u + v \leq 3$, the reverse triangle inequality in $\mathbb{R}^3$ gives

$$\left(K_j(u)^2 + K_j(v)^2 + K_j(u+v)^2 \right)^{1/2} \geq \sqrt{\delta} - \sqrt{3} e_j.$$

Choose j sufficiently large that $\sqrt{3} e_j < \sqrt{\delta}$. Only after this choice may we square to obtain a positive triple bound with

$$\delta_j = \left(\sqrt{\delta} - \sqrt{3} e_j \right)^2 \rightarrow \delta.$$

Put

$$b_j = \frac{2n+1-C_j}{n(n+1)}, \quad q_j = \frac{b_j}{3} - \frac{2}{9n}, \quad a_j = 2\delta_j q_j, \quad B_j = (2n+6)\sqrt{C_j}.$$

For all sufficiently large j , we have $q_j > 0$ and $0 < \delta_j < 1$. Fix such a j before letting T tend to infinity. Write E_j for its finite energy, abbreviate $N = N_K(T)$ and $D = D_K(T)$, and put $\Delta_{n,j} = E_j - (2n+1)N + n(n+1)D$. The unweighted energy formula of Section 4 and Lemma 2 give $E_j/N = C_j + o_j(1)$ and $D/N \geq b_j + o_j(1)$, where throughout this step $o_j(1)$ denotes a quantity tending to zero as $T \rightarrow \infty$ with j fixed. Apply Steps 2–4 to this same smooth kernel, with $d = \delta_j$. With $\kappa = N_C/N \in [0, 1]$, drop the positive final term in (37) to get

$$\frac{\Delta_{n,j}}{N} \geq 2\delta_j \left(\frac{D}{3N} - \frac{2Y_0}{9N} - \frac{2}{3N} \right) - (2n+6)\sqrt{\kappa E_j/N} \geq a_j - B_j\sqrt{\kappa} + o_j(1).$$

Together with the exact defect bound $\Delta_{n,j}/N \geq \kappa$, this yields

$$\frac{\Delta_{n,j}}{N} \geq \max\{\kappa, a_j - B_j\sqrt{\kappa}\} + o_j(1).$$

The error here is uniform in $\kappa \in [0, 1]$; for example,

$$\left| \sqrt{\kappa E_j/N} - \sqrt{\kappa C_j} \right| \leq \left| \sqrt{E_j/N} - \sqrt{C_j} \right| = o_j(1).$$

The first branch κ is increasing and the second branch $a_j - B_j\sqrt{\kappa}$ is decreasing. Since $a_j, B_j > 0$, they meet once on $\kappa \geq 0$. Writing $y = \sqrt{\kappa}$, the intersection solves $y^2 + B_j y - a_j = 0$ and has $y = 2a_j / (\sqrt{B_j^2 + 4a_j} + B_j)$. Thus

$$\min_{\kappa \geq 0} \max\{\kappa, a_j - B_j\sqrt{\kappa}\} = \left(\frac{2a_j}{\sqrt{B_j^2 + 4a_j} + B_j} \right)^2.$$

Minimizing on $[0, \infty)$ instead of the actual range $[0, 1]$ can only weaken the lower bound. Finally the defining identity for $\Delta_{n,j}$ gives

$$\frac{D}{N} = \frac{2n+1 - E_j/N}{n(n+1)} + \frac{\Delta_{n,j}/N}{n(n+1)} = b_j + \frac{\Delta_{n,j}/N}{n(n+1)} + o_j(1).$$

Consequently, for every such fixed j ,

$$\liminf_{T \rightarrow \infty} \frac{D_K(T)}{N_K(T)} \geq b_j + \frac{1}{n(n+1)} \left(\frac{2a_j}{\sqrt{B_j^2 + 4a_j} + B_j} \right)^2.$$

Now let $j \rightarrow \infty$ on the right. Continuity at (C, b_0, q_0, δ) proves (33). This order of limits asserts no asymptotic for the energy of the nonsmooth endpoint profile and requires no endpoint-support pair theorem. $\square$

This explicit improvement is small. Its role is to show that retaining height-window geometry can go strictly beyond the optimized common-kernel moment certificate, while controlling off-line zeros without GRH. It is a fixed-field asymptotic statement: it supplies neither an explicit height from which the improvement is visible nor a uniform error as the field varies.

6 Comparison with earlier bounds

When $K/\mathbb{Q}$ is Galois, the Aramata–Brauer theorem [3, 7] makes ζ_K/ζ entire. Hence each zero of ζ occurs in ζ_K with at least the same multiplicity, and $N_K(T) \sim nN_\zeta(T)$. Lamzouri’s bound [11, Theorem 1.1] therefore gives the inherited estimate $0.8362503518\dots/n$. (For a general non-Galois field, the corresponding divisibility statement is Dedekind’s conjecture and is not used here.) The degree-only specialization of Theorem 1 improves the inherited Galois estimate for every $n \geq 2$, and applies to non-Galois fields as well. In low degrees the numerical comparison is:

Degree	Galois inherited bound	Theorem 1	Best cited GRH bound
2	0.4181	0.4449	0.4585
3	0.2788	0.3057	0.3064
4	0.2091	0.2334	0.2338

The quadratic GRH figure is from [9]; the degree-three and degree-four figures follow from the squared-multiplicity estimates reported in [15]. See also [2] for Galois fields. The new estimate applies unconditionally to every fixed number field and retains the permutation-rank parameter r , although the displayed GRH figures are slightly larger. It concerns distinct complex locations in the whole critical strip, not simple zeros or zeros on the critical line, and is neither uniform in the field nor effective in the starting height.

A secondary abelian consequence. If $K/\mathbb{Q}$ is a fixed finite abelian extension, define

$$N_{K,0}(T) = \sum_{\substack{\rho \in \mathcal{Z}_K(T) \\ \operatorname{Re} \rho = 1/2}} m_\rho.$$

Then

$$\liminf_{T \rightarrow \infty} \frac{N_{K,0}(T)}{N_K(T)} \geq \frac{3}{2} - \frac{1}{\sqrt{2}} \cot(1/\sqrt{2}) = 0.67250070367941164 \dots$$

Indeed, class field theory gives $\zeta_K(s) = \prod_{\chi \in X_K} L(s, \chi)$, where X_K is the finite set of primitive Dirichlet characters attached to K . Apply [1, Theorems A–B and Remark 6.1] to each factor. For every $\varepsilon > 0$, all factorwise inequalities hold simultaneously once T is sufficiently large, because X_K is finite. Orders of vanishing add, even at common zeros, so summing and then letting $\varepsilon \downarrow 0$ proves the displayed inequality. This counts multiplicity and gives no assertion that a zero of the product is simple.

Use of AI: The author prepared a 6-page draft (without using AI) with outlines of proofs extending Lamzouri’s method to Dedekind zeta functions. The author then fed this draft to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) to work out the details of the equations and identify missing gaps. The author fixed those missing gaps and iteratively verified and expanded the proofs using AI tools. The author is solely responsible for the correctness of the proofs.

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